DATA SCIENCE WITH PYTHON LOG SHEET – 30 DAYS

DAY	DATE	TOPIC	KEY ACTIVITIES COMPLETED	SKILLS GAINED	FACULTY SIGNATURE
1		Introduction to Data Science	Discussed data science role & real-world applications	Understanding DS fundamentals	
2		Python Installation & Setup	Installed Python 3 and configured environment	Environment setup skills	
3		Python IDEs	Explored VS Code, Jupyter Notebook, Google Colab	IDE handling	
4		Python Variables & Data Types	Practiced variables and data types	Basic programming foundation	
5		Control Flow Statements	Implemented conditional statements	Logical decision making	
6		Loops in Python	Practiced for and while loops	Iteration & automation	
7		Python Functions	Created functions with arguments & return values	Modular coding	
8		Exception Handling	Implemented try-except blocks	Error handling	
9		Python Data Structures	Worked with strings, lists, dictionaries, tuples	Data storage skills	
10		NumPy Basics	Performed numerical operations using arrays	Numerical computing	
11		Pandas Basics	Created Series & DataFrames	Data manipulation	
12		Loading Data from Files	Imported CSV, Excel, JSON files	Multi-format data handling	

13		SQL Data Loading	Loaded SQL database data into DataFrames	Database integration	
14		MongoDB Integration	Loaded MongoDB data into Pandas	NoSQL handling	
15		Web Scraping	Extracted webpage data using Beautiful Soup	Data extraction	
16		Matplotlib Visualization	Created line charts, bar plots, histograms	Basic visualization	
17		Seaborn Visualization	Built pair plots, count plots, violin plots	Statistical visualization	
18		Plotly Visualization	Created interactive scatter & bar charts	Interactive dashboards	
19		Project – I Planning	Dataset exploration & workflow planning	Project structuring	
20		Data Preprocessing	Identified data quality issues	Data preparation	
21		Handling Missing Data	Treated null values using imputation	Data cleaning accuracy	
22		Data Aggregation & Grouping	Performed grouping and summary statistics	Analytical skills	
23		Data Transformation	Applied scaling & normalization	Model-ready data preparation	
24		Feature Engineering	Created new features	Improved model input	
25		Encoding Categorical Data	Applied Label & One-Hot Encoding	ML compatibility	
26		Handling Outliers	Detected anomalies & treated outliers	Bias reduction	
27		Handling Imbalanced Data	Applied sampling techniques	Dataset balancing	
28		Exploratory Data Analysis	Conducted univariate & bivariate analysis	Insight generation	

29		Hypothesis Testing & Statistical Tests	Performed T-Test, Z-Test, ANOVA, Mann-Whitney	Statistical decision making	
30		Final Visualization & Project Completion	Completed analysis & presented findings	Professional presentation skills	